Design Documentation

Introduction

The **Smart Route Planning** project is a full-stack web application built to solve complex route optimization problems for field service scheduling. The system is designed for performance, flexibility, and a modern developer experience. This document outlines the complete technology stack used in both the frontend and backend, along with a high-level overview of the architecture and components.

Frontend Stack

Technology	**Description**
TypeScript	Strongly typed programming language that builds on JavaScript to ensure better developer tooling and fewer runtime errors.
ReactJS	Component-based frontend library for building interactive user interfaces.
Tailwind CSS	Utility-first CSS framework enabling rapid UI development through composable utility classes.
Shadcn UI	Accessible and customizable component library built on Tailwind CSS. (Replaces JoyUI from initial plan.)
Tanstack Router	Type-safe, modern routing library for managing page navigation in React applications.
Tanstack Query	State-of-the-art data-fetching and caching library for synchronizing server state with the UI.
Google Maps API	Provides powerful geospatial features including interactive maps and route rendering.

Frontend

The frontend of the application is built using Shadcn UI for accessible, consistent components and styled with Tailwind CSS. The layout includes key components such as a Map View (for visualizing routes), Schedule View (daily plans), Worker Details (individual schedules), and a Solution Comparison tool. Navigation is handled with Tanstack Router for type-safe routing. Data fetching and caching are managed with React Query, with local state handling for UI preferences and input validation via an Input Analyzer. The Google Maps integration enables route visualization, map controls, and geocoding for address-to-coordinate conversion.

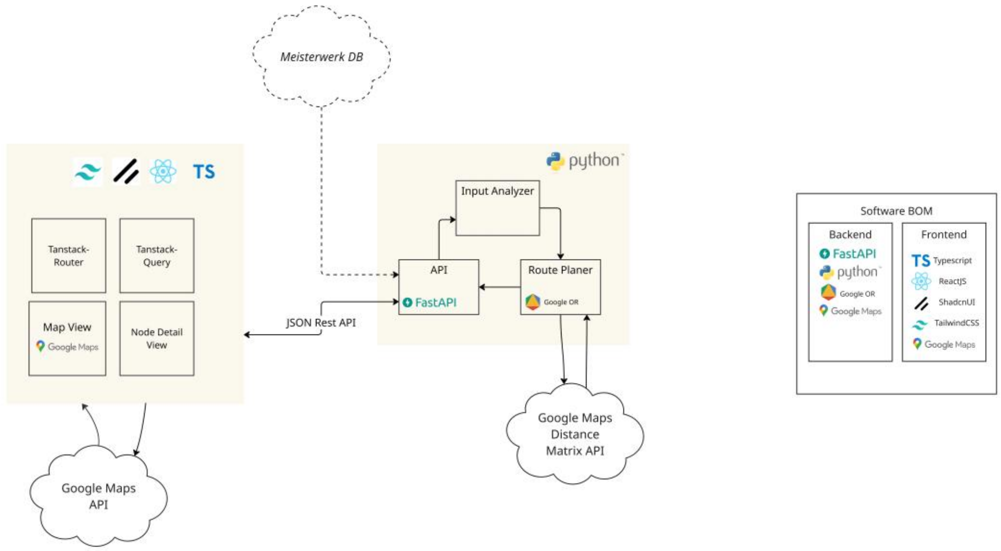
Backend Stack

Technology	Description
Python	General-purpose programming language used for backend development and algorithm integration.
FastAPI	High-performance web framework that provides asynchronous API endpoints with automatic validation and OpenAPI generation.
Google OR-Tools	Powerful combinatorial optimization toolkit used to generate optimal routes under complex constraints.
Google Maps Distance Matrix API	External service used to calculate travel distances and durations between addresses.

Backend

The backend is built with FastAPI to provide high-performance RESTful endpoints. Core business logic is organized into service classes, data models for validation, and shared utilities. The optimization engine uses Google OR-Tools with separate solvers for fixed and flexible scheduling, supported by a constraints handler and a route calculator. External services include a Google Maps API client for distance matrix calculations and modules for importing/exporting JSON data.

Architecture Diagram



Summary

Smart Route Planning combines a modern React frontend with a high-performance Python backend to deliver an efficient, user-friendly system for route planning and optimization. The use of proven libraries such as OR-Tools and FastAPI ensures both accuracy and speed in generating results, while the frontend stack emphasizes maintainability and interactivity through type safety, modular design, and stateful UIs.